

TD-ML: A Machine Learning Toolkit for Real-Time Audiovisual Performance and Installation

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Abstract

*This paper presents **TD-ML**, a modular machine-learning toolkit for **TouchDesigner**, a node-based visual programming environment for rapid real-time media authoring. **TD-ML** addresses the lack of trainable in-graph machine learning in **TouchDesigner** by providing a suite of high-level operators for supervised and unsupervised methods, including classification, regression, clustering, and dimensionality reduction. These operators offer interpretable, adaptable tools that support exploratory, iterative design. Unlike frameworks centered on large-scale deep learning, **TD-ML** prioritizes lightweight, CPU-based models that can be readily trained on custom datasets and be executed with very fast inference times, also on low-cost computers. By integrating these capabilities directly into **TouchDesigner**'s visual dataflow paradigm, **TD-ML** enables users to build adaptive systems that respond dynamically to multimodal inputs such as video, audio or sensor data without the need for extensive programming knowledge. The design emphasizes interpretability and authoring speed while preserving fine-grained control over training, feature selection, and parameterization, enhancing opportunities for creative expression and domain-specific customization. By reducing setup overhead, it supports rapid prototyping, and makes controllable ML techniques practical for computational media practitioners. Exemplary use cases are presented that span gesture and audio recognition, responsive audiovisual composition, and adaptive interaction.*

1. Introduction

Recent discourse in machine learning is dominated by the building of ever-larger foundation models. However, a parallel paradigm - small-data, efficient, interactive machine learning (IML) — offers distinct benefits for creative practice. Working with compact, adaptive models lowers resource demands, enables real-time operation on affordable hardware, and foregrounds individual expression. Previous work shows that personalized, smaller models can help restore character and artistic integrity in AI-mediated creativity [BKSF*25]. In performance contexts, dynamic adaptation to gestures, environments, and multimodal streams yields expressive, interactive experiences that are unavailable from static, offline pipelines.

IML has emerged as a creative medium in its own right: artists and designers use it as an exploratory, example-driven process rather than a one-shot optimization. This practice emphasizes situated data, rapid iteration, and human steering. It is well documented in the NIME community through examples such as gesture-sound mapping, real-time retraining, and embedded workflows that avoid cloud-scale infrastructure [JC23, SBF17]. These workflows echo the core HCI principles – rapid and reversible feedback, transparency of system status, and co-adaptation of the user with the model [FO03]. Despite the recognized value of IML, the imple-

mentation of these techniques often requires substantial programming and ML expertise, leaving many creatives behind.

TouchDesigner (TD) is a popular node-based platform for real-time media. However, it lacks a native, high-level toolkit to enable non-experts to capture examples, train lightweight models, and refine mappings entirely within TD. Existing options either live outside TD and communicate via OSC (Open Sound Control) — for example, **Wekinator** or **FluCoMa** (for **Max/MSP**, **Pure Data**, **SuperCollider**) — or they expose low-level ML primitives that require expertise (e.g., **TDNeuron**), or provide only narrow, one-off integrations. To address this gap, we introduce **TD-ML**, a novel toolkit that integrates IML capabilities natively and seamlessly into **TouchDesigner**. **TD-ML** offers user-friendly high-level operators with an interface to record and label examples directly from **TouchDesigner**'s **CHOP/TOP/DAT** pipelines (**CHOP** for Channel Operator, **TOP** for Texture Operator, **DAT** for Data Operator). It enables users to train and retrain classification, regression, and mapping models in real time, route and monitor model outputs as standard TD nodes. Operating fully inside TD, **TD-ML** eliminates OSC latency and cross-application setup, enabling rapid, in-network iteration. To our knowledge, this is the first end-to-end, high-level IML workflow in **TouchDesigner** designed for non-experts.

Our contributions aim to close the gap between IML's creative promise (as articulated in NIME literature) and the day-to-day

needs of TD practitioners, offering an artist-centered alternative to the scale-at-all-costs AI approach within the TD ecosystem. We first review related creative ML tools in Section 2, then detail TD-ML’s design and implementation in Section 3, and finally present example use cases in live audiovisual performance and installation contexts in Section 4. We discuss the trade-offs of our approach in Section 5 and conclude in Section 6.

2. Related Work

A growing number of toolkits have been developed to lower the barrier to using machine learning in creative practice, particularly for real-time, interactive workflows for artists and designers. Despite differing host environments and audiences, these systems share common goals. They aim to abstract away algorithmic complexity while preserving expressive, low-latency control; they support iterative, on-the-fly training; and they enable fluid mappings between embodied inputs (e.g. gestures) and multimodal outputs (e.g. sound or visuals).

Wekinator provides a graphical user interface for collecting examples and training models in real time. It popularized rapid supervised mapping techniques such as gesture-to-sound or gesture-to-visual control, and it pioneered iterative live training loops that have influenced subsequent interactive ML tools ([FC10, SPF11]). FluCoMa integrates advanced signal processing and machine listening with machine learning into environments like Max/MSP, Pure Data, and SuperCollider. This toolkit allows “techno-fluent” artists to combine signal decomposition, feature extraction, and learning within familiar creative coding environments ([TGR*22]). InteractML brings node-based interactive ML to game engines and virtual reality (VR), enabling creators to configure and train movement classifiers and regressors through a visual programming interface ([HPGD*21]). Together, these platforms embody a shared ethos: empowering creators to leverage ML without deep coding or mathematical expertise.

Within TouchDesigner, TDNeuron is the closest prior effort. It is framed as a deep learning framework with native, low-level integration (using GLSL shaders) and includes educational examples inside TD. However, its emphasis on low-level, GPU-centric deep learning contrasts with our high-level, CPU-oriented toolkit approach ([GB]). MergingEntities explores an interactive audio/visual environment where movement data is mapped via a GPU-based deep neural network to drive sound synthesis, but it exists as a research prototype rather than a general-purpose toolkit ([Sch]).

TD-ML builds on the ideas from these projects by bringing similar interactive ML capabilities directly into TouchDesigner, with an emphasis on modularity and real-time performance for rapid, accessible workflows. In contrast to GPU-centric or low-level frameworks, TD-ML adopts a lightweight, high-level, CPU-based approach. This design prioritizes quick iteration, live training, and expressive input–output mappings, fitting the needs of creators working within TD’s visual programming paradigm.

3. Design and Implementation

TD-ML is implemented as a modular toolkit composed of custom TouchDesigner components (COMPs) packaged as .tox plugins. Built on TouchDesigner’s built-in Python integration, it embeds machine learning models from libraries such as scikit-learn and umap-learn directly into the visual dataflow. Each component exposes its configuration through TouchDesigner’s native parameter interface (as shown in Figure 1), which supports fast creative iteration as well as fine-grained experimental control. The parameters are annotated with descriptive help text sourced from the scikit-learn documentation to aid users in understanding each setting. TD-ML is engineered for seamless use of scikit-learn within TouchDesigner’s CHOP (Channel Operator) ecosystem, relying on Script CHOPs to convert between TouchDesigner’s real-time channel data and NumPy arrays. This design enables real-time signal streams—such as sensor inputs, audio signals, or other interactive control data—to drive end-to-end machine learning workflows entirely inside TouchDesigner.

3.1. Machine Learning Components

TD-ML includes dedicated COMPs for supervised and unsupervised learning:

MLP-Classifer: A multi-layer perceptron neural network for classification tasks. The component exposes nearly all relevant scikit-learn parameters through TouchDesigner’s parameter UI, allowing users to tune model complexity (e.g. number of hidden layers and neurons), activation functions, training algorithms, and regularization settings.

MLP-Regressor: A multi-layer perceptron neural network for regression tasks (predicting continuous values). It similarly provides comprehensive parameter control, enabling experimentation with learning rate, network architecture (hidden layers), and optimization strategies to fit continuous mapping problems.

t-SNE (t-distributed Stochastic Neighbor Embedding): A component for dimensionality reduction, useful for visualizing high-dimensional data in two or three dimensions. The user can adjust parameters like perplexity, learning rate, and number of iterations in real time, enabling dynamic exploration of relationships in the data.

UMAP (Uniform Manifold Approximation and Projection): Another dimensionality reduction tool that seeks to preserve both global and local structure of high-dimensional data when embedding it into a lower-dimensional space. The UMAP component offers full access to parameters such as the number of neighbors, distance metric, and embedding dimensionality, enabling sophisticated analysis and interactive visualization of complex datasets (as demonstrated in Figure 1).

K-Means: A clustering component for unsupervised learning, which groups similar data points into clusters. Users can dynamically adjust the number of clusters K and initialization methods via the TouchDesigner interface, making it easy to experiment with different clustering configurations in real time.

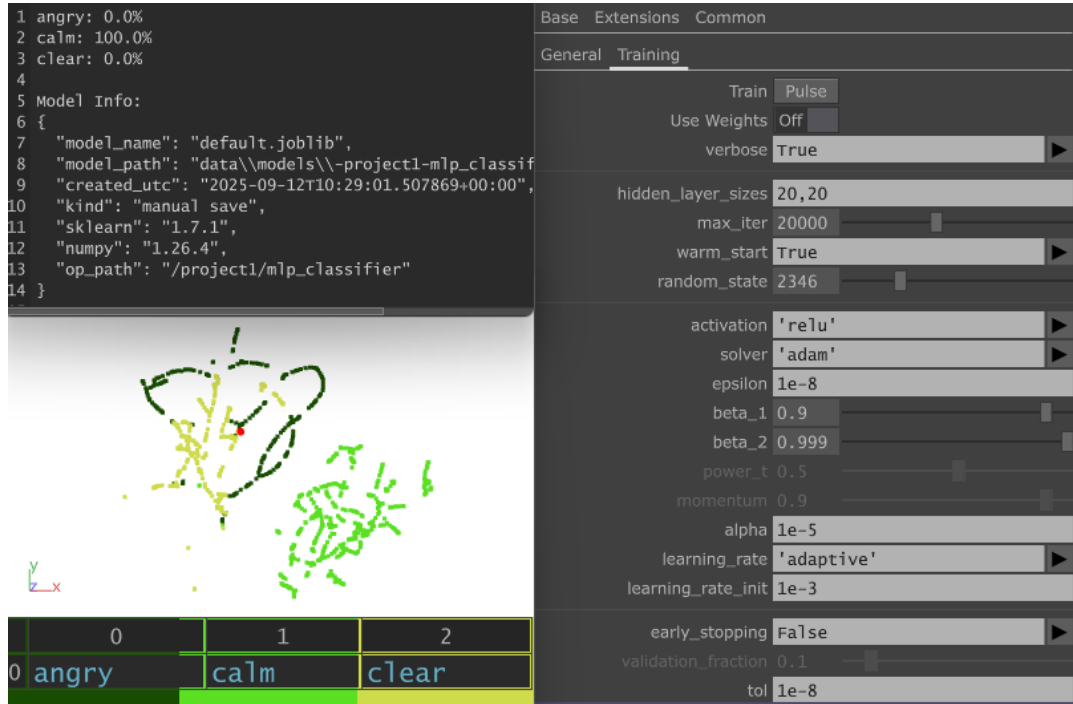


Figure 1: TDML - Classifier Output, UMAP Visualization and Parameter Interface

3.2. Audio Processing Components

MFCC: This component extracts Mel-Frequency Cepstral Coefficients (MFCCs) from incoming audio streams. MFCCs are critical features for audio classification and analysis tasks (for example, they are commonly used to characterize timbre in voice or musical instrument sounds). By providing an MFCC extractor directly inside TouchDesigner's visual environment, TD-ML simplifies the process of audio feature extraction for use in real-time learning workflows.

3.3. Data Management Components

To support rapid iteration and experimentation, TD-ML provides intuitive tools for generating and managing datasets directly from live signal streams:

Datasetter: Streamlines the creation of datasets from CHOP channels, DATs and SOPs, ensuring compatibility for neural network training. Key features include:

- Automated checking of X (input) and y (output) data validity
- Dataset saving and loading.
- Automatic point addition based on user-defined thresholds, enabling quick and intuitive dataset population from real-time signals (sensor, control, or audio)
- data type agnostic input system (CHOP, DAT, SOP)

3.4. Environment and Compatibility Components

TD-ML relies on external Python libraries like scikit-learn, umap-learn, and pandas, which are not included in TouchDesigner's de-

fault Python distribution. Recognizing the importance of reproducibility and easy setup, the toolkit provides mechanisms to simplify environment configuration and dependency management:

Conda-Environment-Loader: This utility component allows users to quickly load a predefined Conda environment from within TouchDesigner. By using a known environment (specified via an environment file or name), it ensures that all required Python dependencies (and compatible versions) are available and consistent across different installations or projects.

Version-Checker: This component automatically verifies the versions of external Python libraries in the current TouchDesigner Python environment. It provides a clear report of the installed versions versus the required versions, helping users catch any incompatibilities or missing dependencies before running the toolkit. This reduces runtime errors during development, deployment, or live performance.

4. Example Use-Cases

We demonstrate TD-ML's capabilities through three example scenarios that represent common real-time creative tasks. Each example is built entirely within TouchDesigner using CPU-based models, and each supports iterative, in-graph retraining with low-latency inference suitable for live performance.

4.1. Example I — Voice Classification from Live Audio

A live audio stream from TouchDesigner's Web Browser component is routed into the MFCC component, which continuously ex-

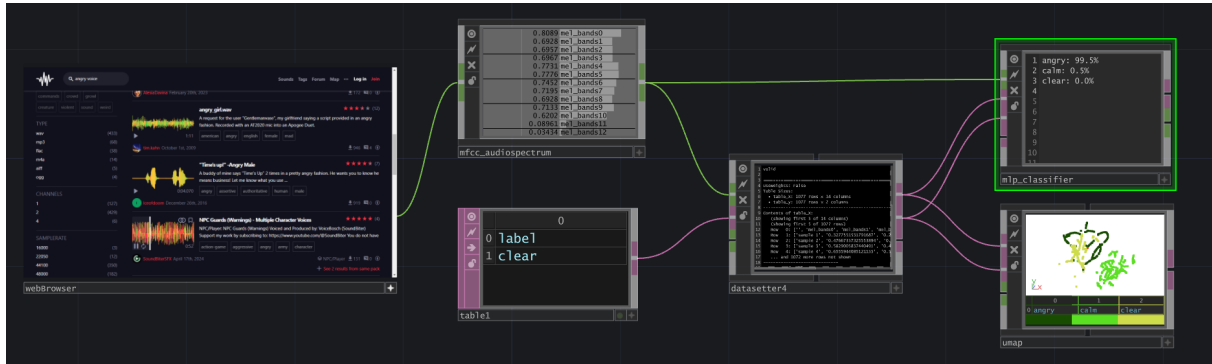


Figure 2: Audio Classification Network trained on vocal samples from freesound.org received directly through TD's Webbrowser

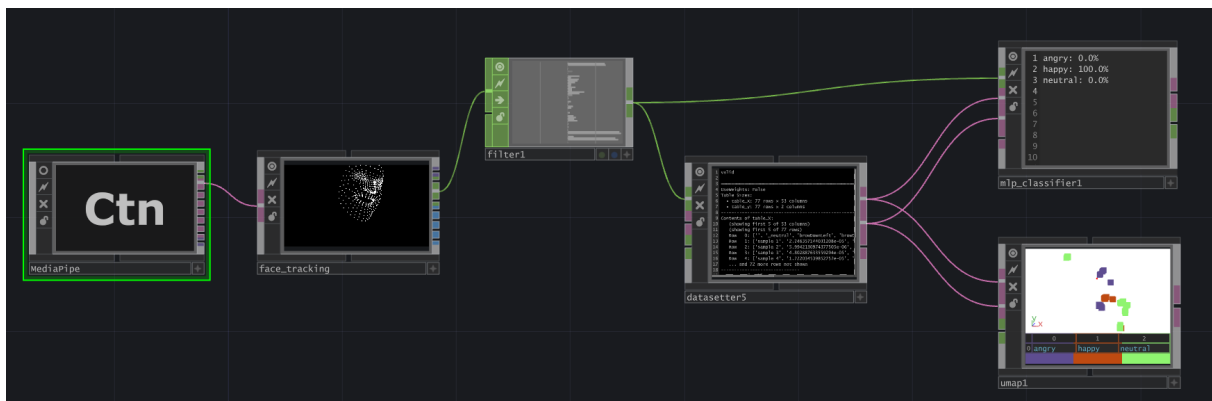


Figure 3: Facial Expression Classification trained on Face Tracking Data through T. Blankensmiths Mediapipe Plugin

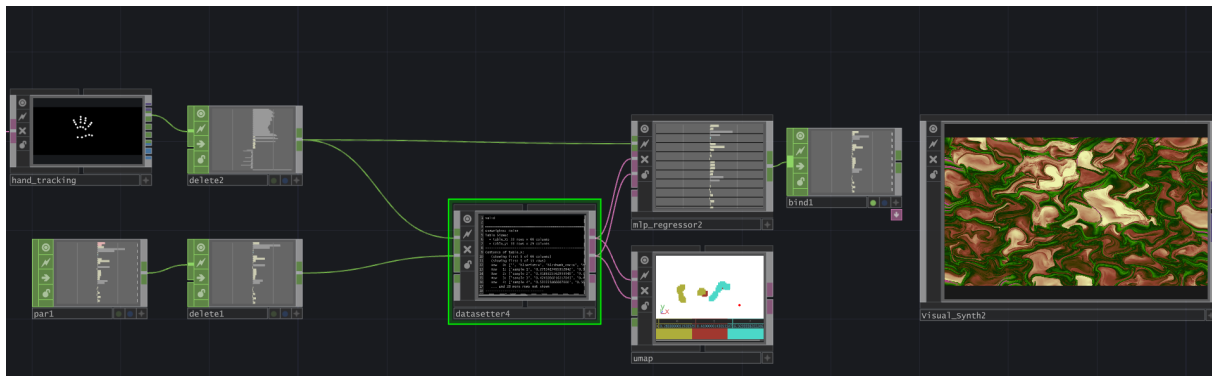


Figure 4: Hand-tracked data (60 parameters) is mapped onto a visual synthesizer (30 parameters) using the MLP Regressor

tracts cepstral feature vectors from the incoming audio. Using the Datasetter, the user assembles a labeled audio dataset directly inside TouchDesigner by capturing live audio examples and labeling them, and by importing additional curated vocal samples (e.g. from freesound.org). Once enough examples are collected for each voice or sound category, an MLP-Classifer is trained in-graph on these examples and then used for frame-wise inference on the live audio stream. The workflow supports rapid addition or removal of training examples and immediate retraining of the model, which enables interactive refinement of the classification boundaries. In practice, this means a creator can quickly adjust the model’s behavior by adding new sounds or re-labeling examples on the fly. Figure 2 shows the resulting TouchDesigner network for this use case and a visualization of the classifier’s output in real time.

4.2. Example II — Facial Expression Classification from Face Tracking

We obtain real-time facial landmark data via Torin Blankensmith’s MediaPipe plugin for TouchDesigner ([Bla]). The raw landmark coordinates (points on the face) are converted into a feature vector representation that is more invariant to head movement — for example, by normalizing coordinates and computing pairwise geometric relationships between key facial points, we reduce the effects of absolute position, scale, and head rotation. Using the Datasetter component, the performer or designer can record example frames for a small set of facial expressions (such as neutral, happy, and surprised), labeling each set of captured landmark features with the corresponding expression. An MLP-Classifer is then trained in real time to recognize these expressions. The classifier’s confidence outputs (for each expression class) are made available as CHOP channels, which can be mapped directly to drive audiovisual controls in the TouchDesigner patch (for instance, to modulate effects or trigger events based on the performer’s expression). This setup allows performers to iteratively refine the training examples during rehearsal and immediately see the impact, while still maintaining a stable, low-latency inference during the actual performance. (See Figure 3 for an illustration of this network and its output.)

4.3. Example III — Nonlinear Gesture-to-Synthesis Mapping via Regression

Hand-tracking features from the same MediaPipe plugin (60-dimensional vectors derived from hand landmark positions) are used to control a visual synthesizer that has 30 adjustable parameters. In this scenario, we use TD-ML’s MLP-Regressor to learn a nonlinear mapping from the high-dimensional hand gesture space to the parameter space of the visual synthesizer. We first normalize the input features (hand tracking data) and the target outputs (synthesizer parameters) within the TouchDesigner network to ensure consistent ranges. Using the Datasetter, the artist can pair specific hand configurations with desired synthesizer parameter settings by example: for instance, record a particular hand pose and concurrently set the visual synthesizer to a certain look or state, and save this as a training pair. After collecting a variety of such gesture-to-parameter examples, the MLP-Regressor is trained on this data. The resulting model learns a compact, nonlinear correspondence

between gestural configurations and synthesizer parameters. This allows for expressive, continuous control of the visual synthesizer through hand movements, without the need to manually design mapping rules. Figure 4 illustrates the end-to-end mapping setup and how the regressor output is integrated into the TouchDesigner network to drive the synthesizer parameters in real time.

5. Discussion

TD-ML demonstrates that interactive machine learning for live audiovisual work is practical without heavy infrastructure. Using lightweight, CPU-based models and in-graph authoring, creators can capture examples, retrain models, and deploy learned mappings directly within a running TouchDesigner patch. By aligning with TouchDesigner’s native CHOP/TOP/DAT dataflow and exposing scikit-learn and UMAP model parameters through familiar UI components, the system lowers the friction of adopting machine learning techniques. It supports an example-centric, iterative workflow that fits naturally into the rapid feedback cycles of rehearsal and performance.

These gains do come with trade-offs. Small, on-the-fly datasets collected within a running patch and on-device training constrain the complexity and generality of models compared to large-scale, GPU-accelerated approaches. Achieving robust generalization beyond the specific conditions rehearsed thus hinges on careful feature engineering (for example, normalizing MFCC features or pre-processing face landmarks as described) and disciplined dataset curation. Moreover, enabling model retraining inside the live environment raises some user-experience risks: without clear feedback, the ability to undo changes, or version control for models, there is a possibility that a model’s behavior could drift or degrade in the middle of a performance.

Future work on TD-ML should address these limitations. We plan to add richer diagnostic tools for model behavior (such as real-time confusion matrices or calibration curves to understand classifier confidence), as well as better dataset management interfaces (e.g. visualizations of class balance and tools for selective example editing or active learning). It would also be valuable to explore optional GPU backends or acceleration for training and inference, and to integrate additional learning algorithms that suit creative contexts (for instance, one-class classifiers for novelty detection or temporal models for sequence data). Benchmarking the system’s latency and throughput within typical TouchDesigner pipelines is another important step to guide optimizations and inform users of performance boundaries.

Finally, conducting user studies with artists and performers will be crucial. Observing how practitioners use TD-ML in real creative scenarios can help evaluate the toolkit’s usability, the speed of the authoring process, the transparency of the system’s behavior, and the perceived expressivity it enables. Such evaluations will inform the design of new features and help balance the desire for immediate, flexible model control against the need for reliability and predictability in live creative settings.

6. Conclusion

We have presented TD-ML, a modular toolkit that brings supervised and unsupervised machine learning techniques into TouchDesigner’s visual programming environment. By prioritizing interpretable, small-data workflows with rapid in-graph retraining, TD-ML makes iterative model building and refinement practical during both the authoring process and live performance. The toolkit provides classification, regression, clustering, and dimensionality reduction components that can be used without leaving TouchDesigner or writing extensive code.

We demonstrated TD-ML’s versatility through three example use cases: an MFCC-based voice classifier, a MediaPipe-based facial expression recognizer, and a nonlinear gesture-to-visuals mapping using an MLP regressor. In each case, the system achieved low-latency, controllable behavior suitable for real-time performance contexts. Taken together, TD-ML operationalizes interactive machine learning principles within everyday TouchDesigner workflows and offers a reproducible, extensible foundation for creating adaptive audiovisual performances and installations.

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